

INHA UNIVERSITY IN TASHKENT

School of Computer and Information Engineering
Fall semester 2022

Practice Problems- 2

Object Oriented Programming MSC 1051

1. A class has 50 students. Every student is supposed to give three examinations. Write a program to read the marks obtained by each student in all three examinations. Calculate and display the total marks and average of each student in the class.
2. Write a program that displays all the numbers from 1 to 100 that are not divisible by 2 as well as by 3.
3. Write a program to calculate the parking charges of a vehicle. Enter the type of vehicle as character (like c for car, b for bus, etc.) and number of hours then calculate the charges as given below:

Truck/Bus---- \$5 /hour

Car-----\$2/Hour

Two wheeler-----\$.50/hour

4. Write a program to print sum of all odd numbers from 1 to 100.
5. Change the following for loop into a while loop. Also convert the for loop into do-while loop.

```
int i;  
for(i=10; i>0 ; i--)  
cout<<i;
```

6. Write a program that accepts any number and prints the number of digits in that number.
7. Write a program to enter a hexadecimal number calculate and display the binary equivalent of this.
8. Write a program to calculate sum of cubes of first n numbers.
 $1^3 + 2^3 + 3^3 + \dots + n^3$
9. Write a program to generate calendar of a month. Input is the start day of the week and the number of the days in the month.
10. WAP to produce following output:

```
A B C D E F G F E D C B A  
A B C D E F   F E D C B A  
A B C D E     E D C B A  
A B C D       D C B A  
A B C         C B A  
A B           B A  
A             A
```


Practice Problems

Fall 2022

Object Oriented Programming-1

1. Write a C++ program to accept two integers and check whether they are equal or not
2. Write a C++ program to check whether a given number is even or odd.
3. Write a C++ program to check whether a given number is positive or negative.
4. Write a C++ program to read the value of an integer m and display the value of n is 1 when m is larger than 0, 0 when m is 0 and -1 when m is less than 0.

Test Data : -5

Expected Output :

The value of n = -1

5. Write a C++ program to accept a coordinate point in a XY coordinate system and determine in which quadrant the coordinate point lies.

Test Data : 7 9

Expected Output :

The coordinate point (7,9) lies in the First quadrant.

6. Write a C++ program to find the eligibility of admission for a professional course based on the following criteria:

Marks in Maths ≥ 65

Marks in Phy ≥ 55

Marks in Chem ≥ 50

Total in all three subject ≥ 180

or

Total in Math and Subjects ≥ 140

7. Write a C++ program to read temperature in centigrade and display a suitable message according to temperature state below :
Temp < 0 then Freezing weather
Temp 0-10 then Very Cold weather
Temp 10-20 then Cold weather
Temp 20-30 then Normal in Temp
Temp 30-40 then Its Hot
Temp ≥ 40 then Its Very Hot
8. Write a C++ program to check whether a triangle is Equilateral, Isosceles or Scalene.
9. Write a C++ program to sum the series

$$1/2 + 2/3 + 3/4 + \dots + n/(n+1)$$

10. Write a program to calculate sum of cubes of first n numbers.

Test Data:

N=5;

$$1^3 + 2^3 + 3^3 + 4^3 + 5^3 = 225$$

11. Write a program to print multiplication table of any number.

Test Data:

Multiplication Table of 3:

$$3*1=3$$

$$3*2=6$$

$$3*3=9$$

$$3*4=12$$

$$3*5=15$$

$$3*6=18$$

$$3*7=21$$

$$3*8=24$$

$$3*9=27$$

$$3*10=30$$

12. Write a program to read a character in uppercase and then print it in lower case.

13. Write a program to calculate the bill amount for an item given its quantity sold, value, discount and tax.

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Practice Problems

Object Oriented Programming MSC 1051

Functions:

1. Write a general purpose function to convert any given year into its roman equivalent. The following table shows the roman equivalents of decimal numbers:

Decimal	Roman
1	I
5	V
10	X
50	L
100	C
500	D
1000	M

Example:

Roman equivalent of 1988 is mdccclxxxviii

2. Any year is entered through the keyboard. Write a function to determine whether the year is a leap year or not.
3. A positive integer is entered through the keyboard. Write a function to obtain the prime factors of this number.

For example, prime factors of 24 are 2, 2,2 and 3, whereas prime factors of 35 are 5 and 7.

Modify the function suitably to obtain prime factors recursively.

4. Write a function which receives a float and an int from main() , finds the product of these two and returns the product which is printed through main().
5. Write a function which receives 5 integers and returns the sum, average and standard deviation of these numbers. Call this function from main() and print the result in main().
6. A 5 digit integer is entered through keyboard, write a function to calculate sum of digits of the 5 digit number:

Without using recursion

Using recursion

7. Write a program using a function that calculates hypotenuse of a right angled triangle.

8. Write a program using functions that enters a number n as input and returns average of numbers from 1 to n.
9. Write a function `_prime` that returns a 1 if the argument passed to it is a prime number otherwise a 0.
10. Write a program to read an integer. Print the reverse of this number using recursion.
11. Write a program using functions to calculate x to the power of y , where y can be either positive or negative.
12. Write a function to draw the following pattern on the screen

```

*****
!      !
!      !
!      !
*****

```

Arrays:

13. Write a program which performs the following tasks:
 - Initialize an integer array of 10 elements in the main()
 - Pass the entire array to a function `modify()`
 - In `modify()` multiply each element of array by 3
 - Return the control to `main()` and print the new array elements in `main()`
14. Write a program to find mean of n numbers using arrays.
15. Write a program to print the position of the smallest of n numbers using arrays.
16. Write a program to find the second biggest number using an array of n numbers.
17. Write a program to enter n number of digits. Form a number using these digits.
18. Write a program to read a 2D array marks which stores the marks of five students in three subjects. Display highest marks in each subject.
19. Write a program to pick up the largest number from any 5 row by 5 column matrix.
20. Write a program to obtain transpose of a 4×4 matrix. The transpose of a matrix is obtained by exchanging the elements of each row with the elements of the corresponding column.

<pre> 1. int main() { char str[] = "C++ Programming"; char *pc; pc = &str[1]; cout << *pc << endl; cout << pc[2] << endl; cout << *(pc + 3) << endl; cout << *(str + 5) << endl; system("Pause"); return 0; } </pre>	<pre> 2. int main() { int a = 2, b = 4, ar[3] = { 5, 8, 9 }; int *p = ar; const int *p1 = &a; int *const p2 = &b; ar++; p1 = &b; p2 = &a; p1++; p2++; p++; *p1 = *p2; *p2 = *p1; *p1 = 10; *p2 = 20; system("Pause"); return 0;} </pre>
<pre> 3. int main() { char st[] = "Final Exams"; char *pc = st; cout << st << endl; for (int i = 0; i < 5; i++) { cout << st[i] << *pc << endl; pc++; } system("Pause"); return 0; } </pre>	<pre> 4. double mean(int x=5,int y=6,int z=7) { return (float)(x + y + z) / 3; } int main() { int p = 3, q = 8, r = 2; cout << mean() << endl; cout << mean(r) << endl; cout << mean(p, q) << endl; cout << mean(p, q, r) << endl; system("Pause"); return 0; } </pre>
<pre> 5. void display(const int[], int); void F1(); void F2(); int main() { F1(); F1(); F2(); F2(); system("Pause"); return 0; } void display(const int a[], int size) { </pre>	<pre> 6. void swap1(int a, int b) { int t = a; a = b; b = t; } void swap2(int &a, int &b) { int t = a; a = b; b = t; } void swap3(int *a, int *b) { int t = *a; *a = *b; </pre>

<pre> for (int i = 0; i < size; i++) { cout << a[i] << endl; } } void F1() { static int array1[4]; display(array1, 4); for (int i = 0; i < 4; i++) { array1[i]++; } } void F2() { int array2[4] = { 11, 12, 13, 14 }; display(array2, 4); for (int i = 0; i < 4; i++) { array2[i]++; } } } </pre>	<pre> *b = t; } int main() { int a = 11, b = 22; swap1(a, b); cout << a << b << endl; swap2(a, b); cout << a << b << endl; swap3(&a, &b); cout << a << b << endl; system("Pause"); return 0; } </pre>
<pre> 7. int main() { int arr[] = { 0, 1, 2, 0, 1, 2, 0, 1, 2 }; cout << arr[3] << endl; cout << arr[arr[3]] << endl; cout << arr[arr[3]+arr[1]] << endl; cout << arr[arr[arr[arr[1]]]] << endl; system("Pause"); return 0; } </pre>	<pre> 8. void func(char a) { cout << a << endl; } int main() { char ch = 256; func(ch); system("Pause"); return 0; } </pre>
<pre> 9. int func() { const int counter = 0; return counter; } int main() { cout << func() << endl; cout << func() << endl; cout << func() << endl; cout << func() << endl; } </pre>	<pre> 10. #include<iostream> using namespace std; int counter = 5; int func() { return counter++; } int main(){ cout << func() << endl; cout << func() << endl; cout << func() << endl; } </pre>

<pre> system("Pause"); return 0; } </pre>	<pre> cout << func() << endl; system("Pause"); return 0; } </pre>
<pre> 11. int main() { int arr[5]; int *ptr; ptr = arr; while (ptr < &arr[5]) { *ptr = ptr - arr; cout << *ptr << endl; ptr++; } system("Pause"); return 0; } </pre>	<pre> 12.int nope(int i) { if (i<=0) return 1; else if (i%2 == 0) return i; else nope(i-3); } int main() { Int x=3; Cout<<nope(x)<<endl; return 0; } </pre>
<pre> 13.#include <iostrem> using namespace std; int func(int i); int main() { int x[4]; int i; for (i=0; i<4; i++) { x[i] = func(i); cout<<x[i]<<endl; } return 0; } int func(int i) { if (i>0) return i + func(i-1); else return 0; } </pre>	<pre> 14. #include<iostream> using namespace std; int func(int); int main() {int a = 2; cout << func(a) << endl; system("Pause"); return 0; } int func(int a) { { ; } { ; } return a; }} </pre>

<pre> 15.#include<iostream> using namespace std; int main() { int arr1[] = { 0, 1, 2, 0, 1, 2, 0, 1, 2, 0 }; int i, arr2[10]; for (i =0; i < 10; i++) { arr2[i] = arr1[9 - i]; } for (i = 0; i < 10; i++) { cout << arr2[i] ; } system("Pause"); return 0; } </pre>	<pre> 16. #include<iostream> using namespace std; int multi(int x, int y) { return (x*y); } int main() { int x=2,y=3,z; z = multi(x, multi(x, y)); cout << z << endl; system("Pause"); return 0; } </pre>
<pre> 17. #include<iostream> using namespace std; void func(int(*parr)[3]); int main() {int arr[2][3] = { 1, 2, 3, 4, 5, 6 }; func(arr); func(arr + 1); system("Pause"); return 0; } void func(int(*parr)[3]) { int i; for (i = 0; i < 2; i++) { cout << (*parr)[i]<<endl; } } </pre>	<pre> 18. #include<iostream> using namespace std; int main() { int arr[] = { 1, 2, 3, 4, 5 }; cout << ++*arr << endl; int *parr; parr= arr + 2; cout << ++*parr - 1 <<endl; cout << 1 + *--parr << endl; system("Pause"); return 0; } </pre>
<pre> 19.#include<iostream> using namespace std; int main() { int num = 5, *ptr = &num; cout << *&num << endl; cout << *&*&num << endl; cout << *&ptr << endl; cout << **&ptr << endl; } </pre>	

<pre>cout << &*&ptr << endl; cout<< *&*&ptr<<endl; system("Pause"); return 0; }</pre>	
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